

# Harshith Kantamneni

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Milwaukee, WI

AI Infrastructure Engineer | GPU/Inference Performance and Agent Systems

## SUMMARY

M.S. ECE engineer working the full AI-systems stack: I optimize CUDA/Triton kernels and profile LLM inference at the bottom, and design agent infrastructure, retrieval, and evaluation on top, measuring everything. Shipped kernel-fusion and GPU-profiling work for LLM workloads, an MCP memory-and-retrieval layer for coding agents, and autonomous multi-agent labs with evaluation and governance.

## SKILLS

- **GPU & Performance:** CUDA, Triton, NVIDIA Nsight Systems/Compute, roofline/bandwidth analysis, quantization, gem5, McPAT
- **AI Systems:** Multi-agent orchestration, MCP server design, Claude Code / Agent SDK, agent evaluation, governance/anti-forgery, LLM inference profiling
- **Retrieval & Memory:** Hybrid search, reciprocal-rank fusion, cross-encoder reranking, sqlite-vec, SQLite FTS5, knowledge graphs
- **Languages & Infra:** Python, C++, CUDA, Rust, TypeScript, Docker, CI/CD, Slurm

## EDUCATION

### University of Wisconsin–Madison

M.S. Electrical and Computer Engineering

Madison, USA

Sep 2024 – Dec 2025

- Relevant Coursework: High Performance Computing, Advanced Computer Architecture, Machine Learning, Fault-Tolerant Computing

### Vellore Institute of Technology

B.Tech in Electronics and Communication Engineering

Amaravati, India

2020 – 2024

## PROJECTS

### bert: Long-Context Memory & Signed-Proof Layer (MCP)

May 2026 – Present

Python, MCP, sqlite-vec, sentence-transformers

- Built **bert**, an **MCP server** (JSON-RPC/stdio) that gives coding agents like Claude Code and Cursor persistent memory, retrieval, and signed proof packets.
- Engineered its hybrid retrieval (dense + BM25 via **reciprocal-rank fusion**, cross-encoder reranking), scoring **nDCG@10 0.6842** on BEIR scifact at **39 ms p50** and **24.9 QPS** on a laptop CPU.
- Added tiered memory and signed in-toto proof packets so agent runs can be reproduced and independently verified.

### Autonomous ML Research Lab (Multi-Agent)

Mar 2026 – Present

Python, C17, Apple Metal, Claude API (Opus 4.8)

- Designed and operate a **30-agent** autonomous research lab on an 18 GB laptop (no cloud) whose agents build and test a from-scratch **C17/Metal training engine** (MoE transformer, 4-bit QAT, hand-written kernels, no ML frameworks) and run pre-registered ML research end to end.
- Built the lab's integrity and governance layer: pre-registration, phase gates, and **anti-forgery** signature verification added after a governance agent forged 5 approvals, with **no recurrence over 23 sessions**.
- Engineered the lab's retrieval substrate: a **4-layer hybrid memory** over **62K chunks** (graph PageRank + sqlite-vec dense + BM25, fused via reciprocal-rank fusion and cross-encoder reranking), warm-served at **~32 ms**.

**HIVE: Autonomous Product Org (Rust + libp2p)**

Mar 2026 – Present

*Rust, libp2p, Python, Claude Code*

- Developed a Rust **Rete** reasoning engine with truth-maintenance and confidence scoring through an autonomous multi-agent workflow, backed by **1,200+** **CI tests** with Miri UB checks and a cargo-deny supply-chain policy.
- Added a **libp2p** P2P node and a retrieval layer (FTS5 + local embeddings) that resolves scoped context from ID-anchored markdown, cutting per-cycle context from **~100K to ~15–20K tokens**.

**Triton vs cuBLAS LLM Kernel Benchmarking**

Dec 2025 – Present

*Triton, CUDA, PyTorch, NVIDIA Nsight*

GitHub

- Benchmarked **76 LLM-shaped** GEMM kernels across **Triton** and **cuBLAS** on an **A100** (p50 latency, throughput, bandwidth, tail jitter), with Triton sustaining **215 TFLOP/s**, **69% of fp16 tensor-core peak**.
- Achieved **up to 1.73×** lower latency than cuBLAS on small-batch FFN projections by fusing linear+bias+GeLU in Triton, cutting **HBM** traffic and kernel-launch overhead.

**Edge LLM GPU Profiling on Jetson Orin Nano**

Feb 2026

*CUDA, NVIDIA Nsight Systems, PyTorch, FlashAttention*

GitHub

- Profiled **attention** and **KV-cache decode** on a Jetson Orin Nano (edge GPU for local LLM-agent deployment), sustaining **~15.4K tokens/sec** single-stream FlashAttention decode and **~7.5 TFLOP/s** peak fp16 GEMM.
- Measured fused **SDPA/FlashAttention** at **~5.6×** over naive attention with Nsight Systems and traced a per-step decode kernel-launch-overhead pattern.

**ML-Guided CUDA Kernel Configuration**

2025

*Python, PyTorch, CUDA, Slurm*

GitHub

- Trained a PyTorch MLP runtime surrogate (12 GEMM/launch features) that predicts CUDA kernel execution time at  **$R^2 = 0.96$**  (MAE 0.018 ms), replacing exhaustive autotuning with argmin-over-grid config selection.
- Selected block/grid configs that ran **within ~3% of the exhaustively-measured optimum** (median across workloads), with benchmarking and CUDA event-timing automated on Slurm.

**MI300X Memory-System Characterization & gem5 Modeling**

2025

*HIP, gem5, McPAT, Python*

- Characterized the **AMD MI300X** memory hierarchy with HIP microbenchmarks (vL1D ~130, L2 ~230, HBM ~630–760-cycle latencies; **4.6–4.8 TB/s** write bandwidth) and calibrated full-system **gem5** to **within 5% MAPE** of measured vL1D latency, down from 89%.
- Compared in-order and out-of-order CPU models in gem5 across SAXPY/DAXPY/IAXPY for CPI, latency, and power, with **McPAT** stats-to-power extraction.